

Schur-functor representations of $USp(2n)$

Symplectic plethysm, polynomial-Gaussian traces, and matrix verification

Proof and verification note

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Outcome. Interpreting $Sp(2n)$ as the compact group $USp(2n)$, the stable formula is the plethystic pushforward of $\text{Exp}_\sigma(e_2)$. Five representative cases in ambient dimensions 2 and 4 give 40 direct comparisons with maximum error 3.70×10^{-15} . A separate Haar $USp(8)$ run detects the invariant alternating form in $\Lambda^2 W$ and verifies odd-degree parity.

1 Convention and stable sigma-MGF

Here $Sp(2n)$ means the compact matrix group

$$USp(2n) = \{g \in U(2n) : g^T J g = J\}, \quad J = \begin{pmatrix} 0 & I_n \\ -I_n & 0 \end{pmatrix}. \quad (1)$$

This convention matters: the complex algebraic group is not a Haar probability space, while its compact form is.

Let $W_n = \mathbb{C}^{2n}$, fix $\lambda \vdash d$, and put $V_{\lambda,n} = S_\lambda(W_n)$. For each partition τ ,

$$[m_\tau] \mathcal{S}_{\lambda,n}^{Sp} = \dim \left(\bigotimes_i \text{Sym}^{\tau_i} S_\lambda(W_n) \right)^{USp(2n)} = \mu_{Sp(2n)}(h_\tau[s_\lambda]). \quad (2)$$

Associativity of plethysm proves the last equality. Since the defining-representation stable sigma-MGF is $\text{Exp}_\sigma(e_2)$, applying its coefficient functional to the fixed function $h_\tau[s_\lambda]$ proves

$$[m_\tau] \mathcal{S}_{\lambda,\infty}^{Sp} = \langle \text{Exp}_\sigma(e_2), h_\tau[s_\lambda] \rangle. \quad (3)$$

A conservative sufficient stable range is $\dim W_n = 2n \geq d|\tau|$, obtained by embedding in tensor degree $d|\tau|$ and applying symplectic Brauer invariant theory.

The central element $-I$ acts by $(-1)^d$ on $S_\lambda(W_n)$. Thus the exact finite-dimensional parity rule is

$$[m_\tau] \mathcal{S}_{\lambda,n}^{Sp} = 0 \quad \text{if } d|\tau| \text{ is odd.} \quad (4)$$

2 Exact trace formula and Gaussian substitution

Functoriality and the Frobenius expansion of s_λ give, for every $g \in USp(2n)$,

$$\text{Tr}(S_\lambda(g)^r) = s_\lambda[X(g^r)] = \sum_{\alpha \vdash d} \frac{\chi^\lambda(\alpha)}{z_\alpha} \prod_{a \in \alpha} \text{Tr}(g^{ra}). \quad (5)$$

This is exact whenever $\ell(\lambda) \leq 2n$.

The joint moment convergence of finitely many symplectic trace powers is an external classical-group trace theorem (Diaconis–Evans, with the normalization below). Let independent $A_k \sim N(0, 1)$ and set

$$Z_k^{Sp} = \sqrt{k} A_k - \mathbf{1}_{2|k}. \quad (6)$$

Joint moment convergence of fixed trace powers, followed by polynomial substitution in (5), gives

$$\mathrm{Tr}(S_\lambda(g)^r) \longrightarrow Q_{\lambda,r}^{Sp} = \sum_{\alpha \vdash d} \frac{\chi^\lambda(\alpha)}{z_\alpha} \prod_{a \in \alpha} Z_{ra}^{Sp}. \quad (7)$$

For $\lambda = (1, 1)$, $Q_{(1,1),1}^{Sp} = \frac{1}{2}((Z_1^{Sp})^2 - Z_2^{Sp})$ has mean 1, representing the invariant symplectic form. For $\lambda = (2)$ the corresponding plus sign yields mean zero.

3 Finite matrix construction and formula cross-check

The sampler performs quaternionic Gram–Schmidt. If v is accepted, it also inserts $-J\bar{v}$; with columns ordered in pairs of blocks this gives both $g^*g = I$ and $g^T J g = J$. The largest observed group residual in all reported sampling is 6.67×10^{-15} .

For the direct route, the code constructs a Young symmetrizer c_λ on $W_n^{\otimes d}$, computes an orthonormal basis B of its image, and forms

$$D_\lambda(g) = B^* g^{\otimes d} B. \quad (8)$$

For the formula route, only $\mathrm{Tr}(g^k)$ is retained. Newton’s recurrence constructs the h_j , and Jacobi–Trudi constructs s_λ . The comparison covers $\mathrm{Tr}(D_\lambda(g)^r)$ for $r = 1, 2, 3$ and the sigma-MGF integrands $\prod_i h_{\tau_i}[D_\lambda(g)]$ for $\tau = (), (1), (2), (1, 1), (3)$.

ambient dimension	tested λ	checks	purpose	status
2	(1)	trace and sigma integrands	rank-one boundary	PASS
4	(2), (1, 1), (3), (2, 1)	trace and sigma integrands	all degree ≤ 3 types	PASS
40 comparisons; maximum absolute error				3.70×10^{-15}

The seeded Haar diagnostic uses 1,200 matrices in $USp(8)$. It estimates means 0.0031 for $\chi_{(2)}$ (target 0), 0.9917 for $\chi_{(1,1)}$ (target 1), and 0.0818 for $\chi_{(3)}$ (target 0). All lie within 4.5 estimated standard errors. The last target also exercises (4).

4 Reproducibility and interpretation

Run the module below from the repository root, or open `symplectic/notebook.py` with marimo. The exact matrix/character identities certify the finite trace formula. The seeded Haar rows are diagnostics only; they neither prove Haar integration identities nor the $n \rightarrow \infty$ convergence theorem.

```
python -m for_this_guy.schur_functor_classical_groups.symplectic.verification
```

References

- S. Howe, *Random matrix statistics and zeroes of L -functions via probability in λ -rings*.
G. Lehrer and R. Zhang, *The Brauer Category and Invariant Theory*.
P. Diaconis and S. N. Evans, *Linear functionals of eigenvalues of random matrices*, Trans. Amer. Math. Soc. 353 (2001), 2615–2633.